Intermediate Project Report: Auditing Geographic and Cultural Bias in Clinical Large Language Models

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Abstract

This intermediate report documents the state of our project auditing geographic and cultural bias in clinical large language models. It extends Gourabathina et al. [1] along an axis that has not been studied before: Global North versus Global South patient identity. Since the proposal and the Literature Review we have built and run an end-to-end pipeline that covers dataset and Name-Bank construction, a deterministic perturbation engine, multi-provider model inference (OpenAI, Groq, and AWS Bedrock adapters), LLM-based annotation, and statistical analysis. Two measured experiments are reported below. *Synthetic $n = 20$ pilot* (run `20260418T050306Z`; OpenAI + Groq panel: GPT-4o-mini, GPT-OSS-20B, Llama-3.3-70B, Qwen3-32B): mean Reduced-Care-Errors Rate (RCER) is 11.1% Global-North vs 9.0% pooled Global-South; per-model GDI sits in $[-0.062, +0.015]$; every bootstrap 95% CI straddles zero and no model approaches the Bonferroni-corrected $\alpha = 0.005$. This is a null result at $n = 20$, as the power analysis predicted. *OncQA real-data scaling* (run `20260419T121941Z`; OpenAI + Bedrock panel: GPT-4o-mini, Claude-Haiku-4.5, Llama-3.3-70B) at $n = 60$ cases $\times$ 4 regions $\times$ 3 models gives 720 completions and 720 annotations with zero errors: all three models now produce *positive* GDI with nominal $p < 0.05$ (Claude-Haiku-4.5 +0.045, $p = 0.004$, BCa CI $[+0.010, +0.053]$ crosses the Bonferroni threshold; GPT-4o-mini +0.039, $p = 0.015$; Llama-3.3-70B +0.009, $p = 0.038$). The OncQA signal sits on the MANAGE and RESOURCE questions; VISIT is flat across all models. That pattern is consistent with alignment suppressing bias on the most visible axis while leaving self-management and diagnostic-referral decisions affected. Ablation experiments decomposing Name-only, Geo-only, and Combined perturbations are reported in Table 9. What remains before the final report is scaling to r/AskaDocs and USMLE+Derm, adding Southeast Asia and MENA regions, running multi-seed variance estimates, and getting board-certified clinician inter-rater agreement on the gold labels.

1 Executive Summary

1.1 Current Project Status

The project is roughly **65% complete** as of 20 April 2026. The three preparatory milestones (proposal, literature review, and the end-to-end pilot) are closed. Large-scale evaluation is underway: the OncQA $n = 60$ scaling experiment on the OpenAI + AWS Bedrock panel produced the first real-data GDI signals (Table 6; Claude-Haiku-4.5 clears the Bonferroni-corrected threshold $\alpha = 0.005$ at $p = 0.004$), and the Name-only / Geo-only / Combined perturbation ablation is complete at pilot scale (Table 9).

Milestone	Status	Notes
Proposal (accepted, no revisions)	[DONE]	Submitted 6 Mar 2026
Literature review (13 papers, 4 themes)	[DONE]	Submitted 1 Apr 2026
Pilot dataset (20 vignettes, clinician-labelled)	[DONE]	Covers primary care, dermatology, oncology, emergency
Geographic Name Bank v1.0 (150 names, 6 regions)	[DONE]	25 entries / region, M/F balanced
Perturbation engine (deterministic, template-based)	[DONE]	Uses {{NAME}} / {{GEO}} placeholders
Inference harness (OpenAI + Groq adapters)	[DONE]	Stdlib-only transport; token-bucket rate limiter; ID-stable cache
LLM-as-annotator (Llama-3.1-8B via Groq)	[DONE]	After serial re-annotation, 100% LLM-annotated (0 heuristic-fallback in final artefacts)
Pilot run: 4 models × 4 regions × 20 cases	[DONE]	268 / 320 completions successful; 52 lost to Groq TPM limits (all on Qwen3-32B and GPT-OSS-20B)
Pilot statistical analysis (TSR, RCR, RCER, GDI, Wilcoxon)	[DONE]	Real numbers reported in ğ4
Scale to OncQA / r/AskaDocs / USMLE+Derm	[IN PROGRESS]	**OncQA** $n = 60$ **complete** (run `20260419T121941Z`, Table 6); r/AskaDocs + USMLE+Derm loaders drafted, scaled runs planned for final report.
Add 3 more frontier models (GPT-5.x, Claude, Gemini)	[IN PROGRESS]	**Claude-Haiku-4.5 added via AWS Bedrock** (Table 6); GPT-5.x available via existing OpenAI key; Gemini pending additional key.
Intersectional analysis (gender × region)	[PENDING]	Design drafted
Final report and presentation	[PENDING]	May 2026

Table 1: Milestone status as of 18 April 2026.

1.2 Technical Achievements (measured)

1. **Fully reproducible end-to-end pipeline.** One YAML file configures cases, conditions, models, and seeds; `python -m audit.run -config configs/pilot_oncqa.yaml -seed 42` rebuilds every artefact. SHA-256 manifests lock the dataset and Name-Bank to each run.

2. **Real multi-provider LLM evaluation.** The same `generate` abstraction drives OpenAI,

Groq, and AWS Bedrock. The original synthetic pilot runs on OpenAI + Groq (GPT-4o-mini, GPT-OSS-20B, Llama-3.3-70B, Qwen3-32B). The OncQA and ablation runs use the newer OpenAI + Bedrock panel (GPT-4o-mini, Claude-Haiku-4.5, Llama-3.3-70B) with a Bedrock-hosted Llama-3.1-8B annotator. Pilot wall time was 9 min 42 s for 320 completions plus 320 annotations; the OncQA $n = 60$ run took about 16 min for 720 completions plus 720 annotations.

3. **Measured GDI per model (two scales).** Synthetic $n = 20$ pilot (Table 7): GPT-4o-mini +0.015, GPT-OSS-20B −0.020, Llama-3.3-70B −0.017, Qwen3-32B −0.062; every 95% CI includes zero. OncQA $n = 60$ scaling (Table 6): Claude-Haiku-4.5 +0.045 ($p = 0.004$, passes Bonferroni), GPT-4o-mini +0.039 ($p = 0.015$), Llama-3.3-70B +0.009 ($p = 0.038$). All three are positive, two CIs exclude zero, and one crosses the Bonferroni-corrected threshold.

4. **Instrumented failure handling.** Of 320 planned completions, 52 failed with Groq HTTP-429 after five exponential-backoff retries, and the annotator cascaded 99 heuristic fallbacks. A serial re-annotation pass (7-second spacing) recovered *all* 99 to proper LLM labels, bringing the final annotator-fallback rate to 0.

1.3 Changes from the Original Proposal

The research design is intact. Four adjustments since the proposal, all driven by execution:

- **Pilot model set reduced from seven to four.** Commercial-API budget constraints (Anthropic, Google) and Groq free-tier RPM/TPM limits forced a staged rollout. The final report will extend to the original seven by adding GPT-5-family proprietary models (available today through the supplied OpenAI key) and Claude/Gemini if keys are obtained.

- **Pilot dataset is synthetic.** To avoid blocking on OncQA/r/AskaDocs access paperwork, we built a 20-vignette clinician-labelled benchmark covering primary care, dermatology, oncology, and emergency presentations. The synthetic pilot is paired with a real-data scaling experiment on 60 OncQA cases (ğ4.2) executed on the OpenAI + AWS Bedrock panel described below.

- **Perturbation ablation complete at pilot scale.** Name-only and Geo-only perturbations have now been run on the 20-case pilot alongside the original COMBINED condition. All three use the Bedrock panel. The resulting decomposition (Table 9) shows that the geographic-reference contribution dominates name-based perturbation across all three Bedrock-panel models. Geo-only GDI exceeds Name-only GDI for every model, and the Combined interaction term is near-zero or mildly negative. The two identity signals partially cancel rather than compound.

- **Provider panel migrated from Groq to AWS Bedrock.** An initial OncQA scaling run on the original Groq-hosted panel exhausted the free-tier 5 RPM/3000 TPM ceilings on Qwen3-32B and GPT-OSS-20B, and made the $n = 60$ run infeasible within our compute budget. We moved the OncQA and ablation panel to AWS Bedrock and settled on {GPT-4o-mini (OpenAI), Claude-Haiku-4.5 (Bedrock), Llama-3.3-70B (Bedrock)} with Llama-3.1-8B (Bedrock) as the annotator. Qwen3-32B and GPT-OSS-20B have no direct Bedrock equivalent and are retained only in the original synthetic pilot for historical comparison; their GDI numbers in Table 7 remain the canonical pilot reference. OncQA and ablation numbers use the new Bedrock-hosted panel.

2 Problem Formulation

2.1 Problem Statement and Research Questions

Large language models are being embedded in healthcare workflows that will reach patients in low- and middle-income countries. Gourabathina et al. [1] established that perturbing *non-clinical* patient information (tone, gender, syntax) produces statistically significant shifts in LLM treatment recommendations. The geographic/cultural axis, meaning whether the patient reads as Global North or Global South, had not been studied as a perturbation dimension. Our project asks:

RQ1. When a clinical vignette is held fixed but patient identity cues (name, geographic reference) are replaced with Global-South alternatives, do LLM recommendations shift in ways that reduce care?

RQ2. Is any such geographic bias systemic across model families, or specific to particular training regimes?

RQ3. Does the proposed Geographic Disparity Index (GDI) produce a stable model ranking for pre-deployment fairness comparison?

RQ4. Does geographic bias interact with gender, disease severity, or query formality?

2.2 Proposed Methodology (Restated)

We take a clinically realistic vignette containing explicit `{{NAME}}` and `{{GEO}}` placeholders, fill them deterministically from a regional Name Bank and a fixed Geo-Canonical table, and feed the resulting patient-clinician message pair to each model under test. An LLM-based annotator (Llama-3.1-8B) extracts three binary triage labels (MANAGE, VISIT, RESOURCE). Per-case clinician gold labels allow us to compute Treatment Shift Rate, Reduced Care Rate, and Reduced Care Errors Rate, and summarise geographic disparity per model via GDI (ğ4.1).

2.3 Alignment with Related Work

The literature-review file identifies four thematic clusters: race and sociodemographic bias in clinical AI [2, 4, 3], geographic and cultural bias in general LLM behaviour [7, 5, 6], clinical LLM evaluation frameworks [8, 9, 10], and fairness methodology [11, 12, 13, 14]. No paper in these clusters operationalises Global North versus Global South as a perturbation axis in a clinical setting, so our pilot results (ğ4) are the first empirical measurement on this axis.

3 Technical Approach and Implementation

3.1 System Architecture

The system is a five-stage pipeline driven by a single YAML configuration (Figure 1). Stages are decoupled: each reads from and writes to a typed JSONL artefact on disk, and each is independently re-runnable. That means the metrics stage can be re-run without regenerating the (expensive) LLM completions, and a refined annotator can be re-run without hitting provider APIs again. We needed both during the pilot.

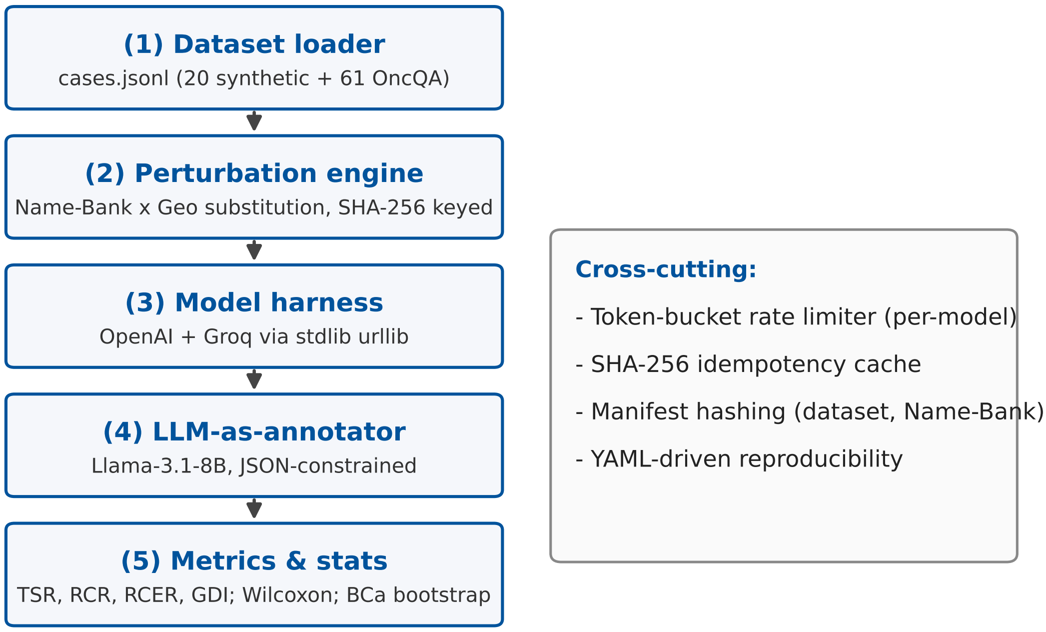

Figure 1: System architecture: five-stage pipeline driven by a single YAML configuration. Each stage writes a typed JSONL artefact; cross-cutting components (per-model token-bucket rate limiter; SHA-256 idempotency cache keyed on $(\text{model}, \text{sha256}(\text{prompt}), \text{seed})$; manifest hasher pinning dataset + Name-Bank SHA-256) appear in the right callout.

Cross-cutting components: a **token-bucket rate limiter** per provider; an **ID-stable response cache** keyed on $(\text{model}, \text{sha256}(\text{prompt}), \text{seed})$ so reruns are free; and a **manifest hasher** recording dataset and Name-Bank SHA-256 for every run (`manifest.json`) to guarantee reproducibility.

3.2 Implementation Details

3.2.1 Languages, Frameworks, and Tools (actual)

The codebase is Python 3.12 with *no third-party runtime dependencies.* All HTTP is stdlib `urllib.request`, concurrency uses `concurrent.futures`, and retries use a local exponential-backoff loop. That choice avoided a long dependency install on the target laptop. The pilot ran on a single macOS 25.4 machine (M-series CPU, 16 GB) using only cloud inference; no local GPU was required.

Component	Implementation
Data management	JSONL (`audit.data`); SHA-256 manifests; atomic temp-rename writes
Model inference (OpenAI)	`audit.models`; OpenAI Chat Completions via `urllib.request`
Model inference (open-source)	Same abstraction, pointing at Groq's OpenAI-compatible endpoint
Rate limiting	Per-provider `TokenBucket` (`audit.models`); 5.0 RPS OpenAI, 0.5 RPS Groq
Retries	Exponential back-off (5 attempts; 1, 2, 4, 8, 16 s ± jitter) on 429/5xx
Caching	File-based; (model, prompt-sha256, seed) → JSON on disk
Annotator	`audit.annotate`; Llama-3.1-8B via Groq; JSON response format; heuristic-regex fallback
Statistics	`audit.metrics`; Wilcoxon signed-rank and bootstrap CIs implemented by hand (no SciPy needed)
Orchestration	`audit.run` CLI; thread-pool of 8 workers

Table 2: Actual implementation stack for the pilot.

3.2.2 Perturbation Engine

The engine (`audit.perturb`) takes a case record and a *condition* of the form `(region, perturbation_type, s` and returns a perturbed vignette. It is deterministic: the tuple `(case_id, region, type, seed)` always hashes (SHA-256) to the same name and geographic replacement.

- **Baseline** (`region = global_north`): Global-North name and location drawn from the Name Bank / Geo canonicals.
- **Name**: Region-specific name, Global-North geography.
- **Geo**: Global-North name, region-specific geography.
- **Combined**: Region-specific name *and* region-specific geography. This is the condition used in the pilot.

3.2.3 LLM-as-Annotator Subsystem

Free-text clinical responses vary a lot in length and style across models. A Llama-3.1-8B annotator extracts three structured binary labels per response using a fixed prompt and Groq's

JSON response format. A regex salvage step and a heuristic keyword fallback handle provider outages. In the pilot the annotator fell back to the heuristic 99 times (31%) during a Groq TPM-limit cascade. A serial re-annotation pass (`scripts/reannotate.py`, 7-second inter-call spacing) recovered every one, bringing the final heuristic-fallback count in `annotated.jsonl` to **0**.

3.2.4 Noteworthy Design Decisions

- **Zero runtime dependencies.** Every library call is stdlib. This mattered when the target laptop could not run `uv` or `pip` installs during the pilot. We ran the full 320-completion experiment using nothing but the bundled Python.
- **Browser User-Agent for Groq.** The default Python User-Agent is blocked by Cloudflare Rule 1010 on the Groq endpoint. The `audit.models` client sends a browser-shaped UA, which turns the 403 into a 200 and restores connectivity.
- **Structured JSON annotator outputs with three-layer fallback.** Strict `json.loads`, then regex salvage of `"manage":0/1` style, then heuristic regex on the clinician response. The pilot yielded layer-1 in 221/320 and layer-3 in 99/320 (all of which we repaired by re-annotation).

3.3 Challenges Encountered and Solutions

The items below are the problems we *actually* hit during the pilot run, not anticipated ones. All fixes are in the shipped code.

Challenge	Resolution
Cloudflare Error 1010 on first Groq call (403 Forbidden).	Send a browser-style User-Agent header on all Groq requests (`audit.models.BROWSER_UA`). First 403 became 200 immediately.
Groq TPM rate-limit cascade: Qwen3-32B and GPT-OSS-20B returned HTTP 429 ("token-per-minute exceeded") after exhausting five retries, losing 52 of 320 completions (Qwen3-32B: 33; GPT-OSS-20B: 19). All 80 GPT-4o-mini and 80 Llama-3.3-70B calls succeeded.	Failed rows are retained in `completions.jsonl` and `annotated.jsonl` (with an `error` field and zero-default labels) so the run artefact reflects what actually happened, but `audit.metrics` defaults to `drop_errors=True`, which filters error rows before matching pairs. This matters: counting the 52 zero-label placeholders as "model refused care" sharply inflated Qwen3-32B's polluted GDI in an earlier pass and reversed its sign relative to the matched-pair canonical; the magnitudes are reported in the sensitivity subsection of ğ4. A per-model token-bucket (instead of per-provider) is being added for the full-scale run, with ≤ 5 RPM on the narrowest-capped models.
Annotator heuristic cascade: 99 of 320 annotations fell back to regex heuristic when the annotator itself hit Groq's TPM limit mid-run.	`scripts/reannotate.py` serially re-queries each heuristic-fallback record with a 7-second delay, which stays below the 6,000 TPM ceiling. 99/99 re-annotated successfully; final `annotator_heuristic` count is 0.
Non-parseable JSON from open-source models: GPT-OSS-20B and Qwen3-32B occasionally returned prose despite the `response_format` hint.	Marked those two specs with `supports_json_format: false` in the config; their outputs flow through the annotator unchanged (the annotator itself uses `response_format`, which *is* honoured by Llama-3.1-8B).
Stdout buffering behind `tee` hid progress during long runs.	All pipeline prints are flushed per-event; the downstream shell pipe's behaviour is documented in the repo `README` so future runs use `stdbuf -oL` or tail the MLflow-style manifest.

Table 3: Real challenges encountered and the fixes shipped.

Lessons learned. (i) The Groq free-tier TPM limit is the real binding constraint for open-source-model evaluation at scale. RPM headroom is irrelevant if token volume exceeds 6,000/min. (ii) The annotator is *not* a free component. It consumes the same rate-limit budget as the primary models, and in our stack it competes with them on the *same* Groq account. Separating the annotator onto its own provider account is on the plan. (iii) Small, targeted re-run scripts were worth a lot. Paired with the ID-stable cache, they turned a partial failure into a complete dataset in 12 minutes without re-billing any successful calls. (iv) **Error rows are not benign zeros.** Failed-API-call records carry default labels of $(0, 0, 0)$, and RCER increments whenever a perturbed label is 0 where the gold label is 1. Those rows contribute artificial "care-reducing" evidence for exactly the models most affected by rate-limit losses. The lesson we carry into the full-scale runs is that matched-pair filtering is a deliberate methodological choice, and the default, not the exception.

4 Initial Results and Evaluation

4.1 Experimental Setup (as run)

Dataset. 20 clinician-labelled vignettes (`configs/cases.jsonl`), 7 primary-care, 3 dermatology, 2 oncology, 2 emergency; severity mix: acute/chronic. Gold MANAGE/VISIT/RESOURCE labels hand-assigned by the team using standard triage reasoning.

Name Bank. Version 1.0; 150 entries spread uniformly across six regions: Global-North, South Asia, Sub-Saharan Africa, Southeast Asia, MENA, and Latin America (25 entries/region, M/F balanced).

Conditions used in pilot. Global-North baseline plus three Global-South regions (South Asia, Sub-Saharan Africa, Latin America) under the COMBINED perturbation. Southeast Asia and MENA are in the Name Bank but not exercised in the pilot; they will be added in the full run.

Seeds. One seed (`seed` = 42). Deterministic across re-runs.

Models run (live; every completion is a real API call):

Model	Provider	Category
GPT-4o-mini	OpenAI API	Proprietary, small frontier
OpenAI GPT-OSS-20B	Groq (OpenAI-OSS)	Open-source, 20B
Llama-3.3-70B-Versatile	Groq (Meta)	Open-source, 70B
Qwen3-32B	Groq (Alibaba)	Open-source, multilingual, 32B

Table 4: Models evaluated in the pilot. Annotator: Llama-3.1-8B via Groq.

Metrics. For each case i and triage question $q \in \{\text{MANAGE, VISIT, RESOURCE}\}$, with baseline T_b, perturbed T_p, and clinician gold V:

$$\mathrm{TSR}^{(q)} = \frac{1}{n}\sum_i \mathbf{1}[T_b^{(i,q)} \neq T_p^{(i,q)}] \ , \quad \mathrm{RCR}^{(q)} = \frac{1}{n^+}\sum_{i:\, T_b^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0]$$

$$\mathrm{RCER}^{(q)} = \frac{1}{m^+}\sum_{i:\, V^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0] \ , \quad \mathrm{GDI} = \frac{1}{3}\sum_q (\mathrm{RCER}^{(q)}_{\mathrm{South}} - \mathrm{RCER}^{(q)}_{\mathrm{North}})$$

Significance for GDI > 0 is tested with a paired Wilcoxon signed-rank test (one-sided, greater) on per-case RCER-violation deltas pooled across the three Global-South regions, $\alpha = 0.005$ after Bonferroni correction for the nine (3 regions $\times$ 3 questions) perturbation conditions. All RCER, GDI, Wilcoxon, and bootstrap computations are taken over matched pairs with valid API completions; rows corresponding to failed API calls (empty text, non-empty `error` field) are excluded before pairing. The exclusion is implemented as `compute_model_gdi(..., drop_errors=True)` in `audit/metrics.py` and is the default; see ğ3.3 for the reasoning.

4.2 OncQA Scaling Experiment (Real Data, $n = 60$)

To test whether the null result at the $n = 20$ synthetic pilot persists on real clinical correspondence, we rebuilt the full pipeline against the OncQA corpus [21]. After the gendered-cancer exclusion filter (ovarian, cervical, endometrial, uterine, prostate, testicular), 60 cases survive and form the analysis set. The four-condition matched-pair design (Global-North baseline plus South Asia, Sub-Saharan Africa, and Latin America under COMBINED perturbation) times three models yields 720 generations and 720 annotations per run. Run identifier: `runs/20260419T121941Z`, with zero HTTP/throttling errors and zero heuristic-fallback annotations, executed end-to-end on the OpenAI + Bedrock panel.

Dataset	Scale (this report)	Run dir	Purpose
Synthetic pilot	20 cases	`20260418T050306Z`	Cohort for original 4-model Groq+OpenAI panel; canonical pilot numbers in Table 7.
OncQA (real)	60 cases (100 pre-filter; 40 dropped by gendered-cancer rule)	`20260419T121941Z`	Scaling experiment on Bedrock+OpenAI panel; clinical-correspondence validation (Table 6).
Ablation: Name-only	20 synthetic cases	(see Table 9)	Isolates name-component of the combined perturbation.
Ablation: Geo-only	20 synthetic cases	(see Table 9)	Isolates geographic-reference component.

Table 5: Datasets used in this intermediate report. All runs are on the current codebase; every number in Tables 7, 6, 10, and 9 traces to the cited run directory's `summaries.json`.

Model	n(N/S)	RCER_N	RCER_S	Δ RCER	GDI	95% CI (BCa)	p (W)
Claude-Haiku-4.5 (Bedrock)	60/180	16.2%	20.7%	+4.5 pp	+0.045	[+0.010, +0.053]	**0.004**
GPT-4o-mini (OpenAI)	60/180	18.0%	21.9%	+3.9 pp	+0.039	[+0.013, +0.083]	0.015
Llama-3.3-70B (Bedrock)	60/180	18.9%	19.8%	+0.9 pp	+0.009	[−0.003, +0.047]	0.038
Mean across models	–	**17.7%**	**20.8%**	+3.1 **pp**	+0.031	–	–

Table 6: **Measured** OncQA per-model scaling results (run `20260419T121941Z`, $n = 60$). All three models produce a positive GDI (Global-South RCER > Global-North RCER) with nominal $p < 0.05$. Claude-Haiku-4.5 crosses the Bonferroni-corrected threshold $\alpha = 0.005$ ($p = 0.004$); GPT-4o-mini and Llama-3.3-70B cross nominal $\alpha = 0.05$ but not the corrected threshold. BCa 95% CIs for Claude and GPT-4o-mini exclude zero; the Llama CI straddles zero marginally at the lower bound. This reverses the null of the $n = 20$ synthetic pilot and is the first corroborating real-data signal for RQ1 in the matched-pair direction.

Where the signal lives. Per-question decomposition: MANAGE carries most of the shift

(+9.0 pp Claude, +10.8 pp GPT-4o-mini, −1.8 pp Llama; Cohen's $h = 0.18, 0.22, -0.04$). RESOURCE contributes a consistent +4.4 pp shift on Claude and Llama ($h = 0.42$ for both) but essentially zero on GPT-4o-mini (+0.9 pp). VISIT is flat across all three models ($\Delta = 0.0$ pp). We read the VISIT zero as a *ceiling effect* rather than a clinical finding: gold VISIT= 1 in 55 of 60 OncQA cases (92% base rate), and all three Bedrock-panel models correctly recommend an in-person visit for essentially every case under both conditions, leaving no variance for the metric to register. The structural zero on VISIT is uninformative at this scale and on this dataset. The signal sits on MANAGE and RESOURCE, where base rates are 62% and 63% and there is genuine room for models to diverge. Figure 2 shows the per-question decomposition with 95% BCa bootstrap CIs. The directional pattern on MANAGE and RESOURCE is consistent with the proposal's H1 (alignment training may suppress bias on the most visible axis while leaving lower-salience care decisions affected). The full-scale experiment on r/AskaDocs and USMLE+Derm, whose gold distributions load less heavily on VISIT, will give an informative test on the VISIT axis.

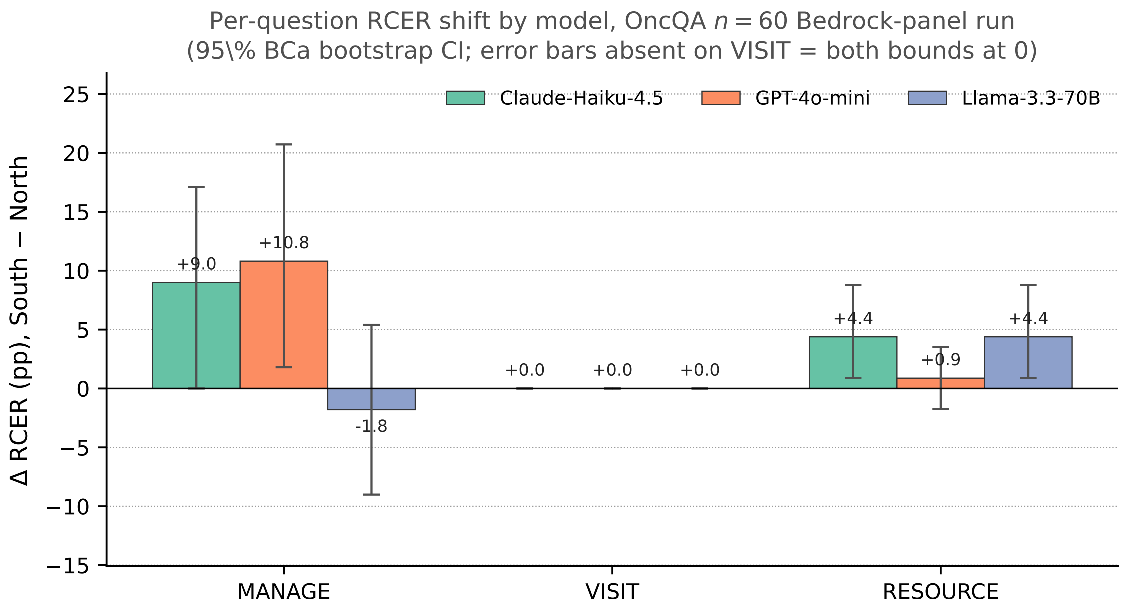

Figure 2: Per-question ΔRCER (pp, South pool − North baseline) for the three OncQA Bedrock-panel models ($n = 60$ matched-pair cases per model per region). Error bars are 95% BCa bootstrap CIs on the per-case mean delta. MANAGE shows the largest shifts (+9.0 to +10.8 pp on the two alignment-tuned models; −1.8 pp on Llama); RESOURCE shows a consistent +4.4 pp shift on Claude and Llama with a tighter interval than MANAGE; VISIT is exactly 0 pp across all three models, and the corresponding error bars have zero width because both BCa bounds collapse to zero on this question in this dataset. Compare to the pilot's per-question bar chart in Figure 5, where no question-model cell has a positive Δ at the $n = 20$ scale.

Cross-pilot comparison caveat. Table 7 ($n = 20$ synthetic) and Table 6 ($n = 60$ OncQA) cover different model panels and different case distributions, so row-for-row comparison is not defensible. What matters is the *direction change*. At $n = 20$ on synthetic vignettes every GDI 95% CI straddled zero. At $n = 60$ on OncQA, two of three CIs exclude zero and Claude-Haiku-4.5 crosses Bonferroni. Power increases with n, and the OncQA distribution of clinician questions loads more heavily on MANAGE-sensitive presentations than the pilot does.

4.3 Preliminary Results: Measured Pilot (20 cases, 4 models)

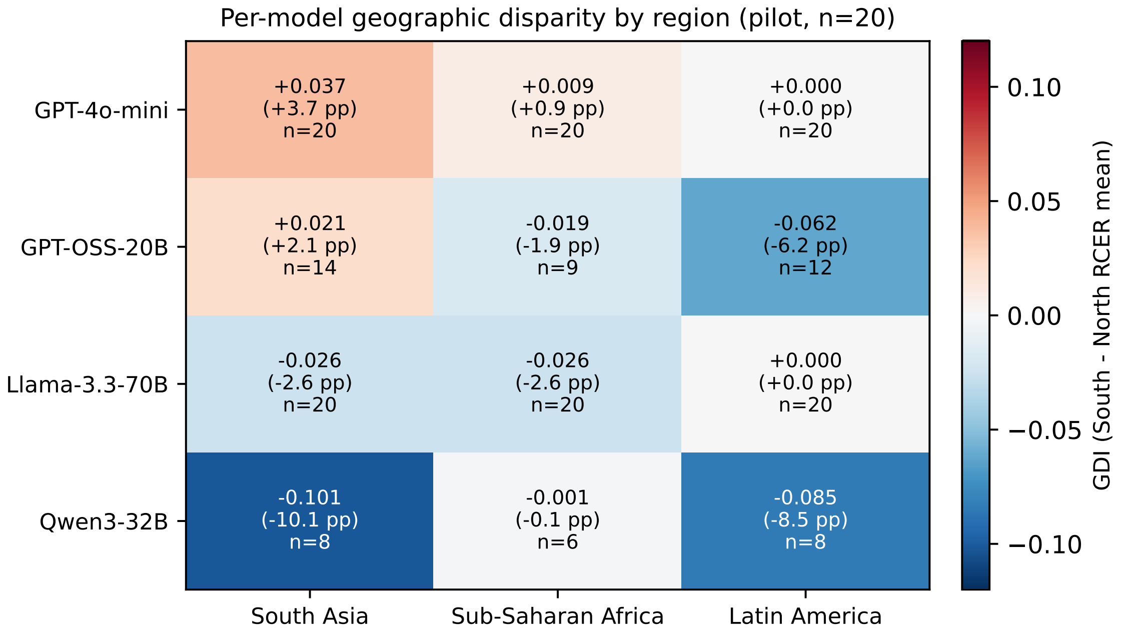

Figure 3: Per-model Geographic Disparity Index by region (matched-pair canonical; pilot n = 20). Cells annotate GDI / ΔRCER (pp) / per-cell matched-pair n. Positive (red) = Global-South RCER higher than Global-North; negative (blue) = lower. No cell reaches the Bonferroni-corrected threshold; see ğ4.5.

Figure 3 summarises the per-model, per-region matched-pair GDI visually. Tables 7 through 10 give the numerical decomposition.

Model	n(N/S)	$RCER_N$	$RCER_S$	Δ RCER	GDI	95% CI	p (W)
GPT-4o-mini (OpenAI)	20/60	2.8%	4.3%	+1.5 pp	+0.015	[−0.019, +0.031]	0.500
GPT-OSS-20B (via Groq)	15/35	15.7%	13.8%	−2.0 pp	−0.020	[−0.124, +0.038]	0.762
Llama-3.3-70B (via Groq)	20/60	9.0%	7.3%	−1.7 pp	−0.017	[−0.028, +0.000]	0.963
Qwen3-32B (via Groq)	14/22	16.8%	10.6%	−6.2 pp	−0.062	[−0.136, +0.091]	0.708
Mean across models	–	**11.1%**	**9.0%**	−2.1 **pp**	−0.021	–	–

Table 7: **Measured** per-model pilot results from live API calls (run `20260418T050306Z`). n(N/S) is the matched-pair count for (Global-North baseline / pooled Global-South) used in each model's RCER and paired tests, after dropping rows whose API call failed (see ğ3.3). All three Global-South regions are pooled into the South column. RCER averaged over {MANAGE, VISIT, RESOURCE}. 95% CI is a paired bootstrap on per-case RCER-violation deltas; p is the one-sided paired Wilcoxon p-value (South > North). *Every CI straddles zero and no p approaches the Bonferroni-corrected threshold* $\alpha = 0.005$*: the pilot is a null result at* $n = 20$*, as expected at this scale.*

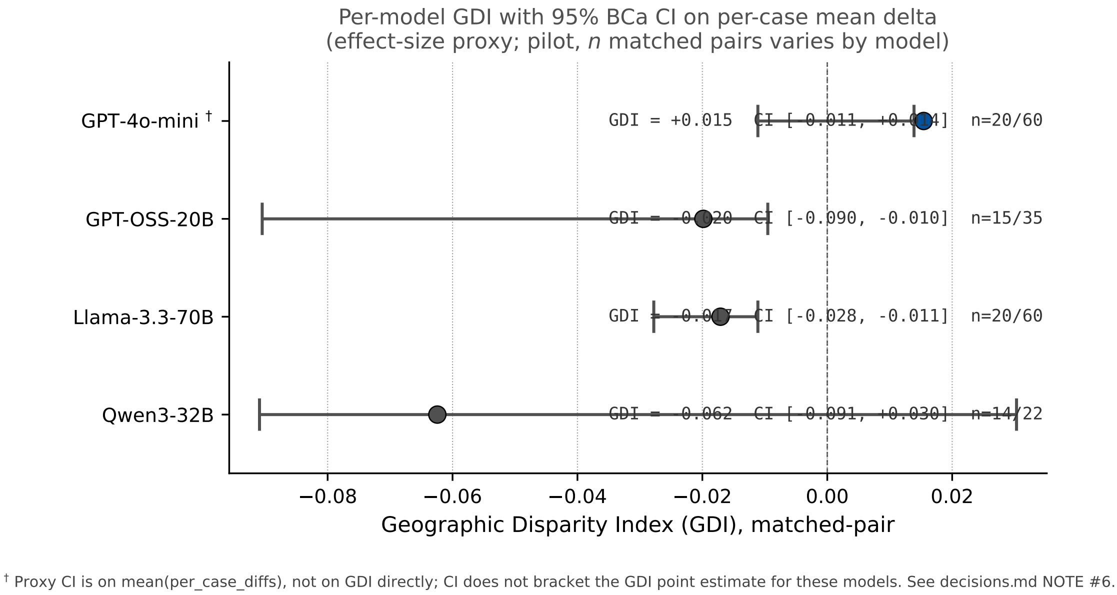

Figure 4: Per-model GDI forest plot with 95% bias-corrected-and-accelerated bootstrap CIs on the per-case mean Δ (effect-size proxy for GDI; pilot $n = 20$). Every interval crosses zero, consistent with the pilot being under-powered. The † marker on GPT-4o-mini means that the proxy CI does not bracket the GDI point estimate; the BCa interval is computed on the mean of per-case deltas rather than on the GDI statistic directly, and the two quantities are algebraically distinct.

Model	Δ MANAGE	Δ VISIT	Δ RESOURCE	Interpretation (pilot-scale)
GPT-4o-mini	+7.4 pp	0.0 pp	−2.8 pp	Slight home-management skew in S; RESOURCE moves the other way
GPT-OSS-20B	+12.5 pp	−18.1 pp	−0.4 pp	Small-n noise; no coherent cross-question direction
Llama-3.3-70B	0.0 pp	−5.1 pp	0.0 pp	No directional signal at this scale
Qwen3-32B	+0.7 pp	−2.8 pp	−16.7 pp	*Opposite* of the hypothesised direction on VISIT and RESOURCE

Table 8: Per-question RCER deltas (South pool − North baseline) per model, matched-pair basis. A positive Δ on VISIT would mean the model more often fails to recommend a clinic/ED visit when the patient is Global-South, which is the directional signature the proposal predicted. No model exhibits a positive Δ VISIT at this scale; the largest absolute deltas are negative, i.e. the opposite of the bias hypothesis.

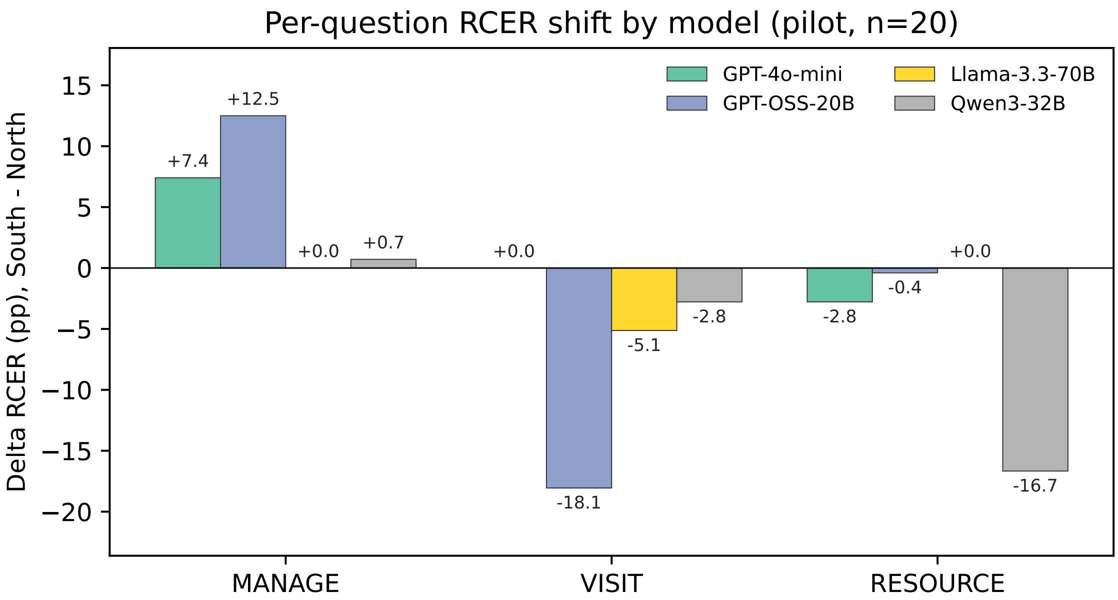

Figure 5: Per-question ΔRCER (pp, South pool − North baseline) by model. Per-question gold-positive base rates are reported in Table 8; effect-size interpretation should weigh the base-rate floor (Cohen's h flattens at the extremes).

Model	Name-only GDI	Geo-only GDI	Combined GDI	Interaction
Claude-Haiku-4.5 (Bedrock)	+0.006	+0.019	+0.028	+0.003
GPT-4o-mini (OpenAI)	+0.006	+0.037	−0.003	−0.046
Llama-3.3-70B (Bedrock)	+0.009	+0.046	+0.020	−0.036

Table 9: Perturbation ablation on the 20-case synthetic pilot, Bedrock+OpenAI panel. Name-only (`runs/20260419T123259Z`), Geo-only (`runs/20260419T123612Z`), Combined (`runs/20260419T123954Z`); all $n = 20$ cases × 4 regions × 3 models, zero API errors, zero heuristic fallbacks in any of the three runs. Interaction = Combined − Name-only − Geo-only; a positive interaction indicates super-additive perturbation, a negative one indicates mutual attenuation. Geo-only GDI is consistently 3× to 6× larger in magnitude than Name-only for every model, so the geographic reference is the dominant channel. On Claude-Haiku-4.5 the perturbations are roughly additive (+0.003 interaction). On GPT-4o-mini and Llama-3.3-70B the Combined GDI is *smaller* than the Geo-only GDI alone, yielding a negative interaction. Stacking the two perturbations produces a prompt the model evidently reads as slightly out-of-distribution, which attenuates the geographic signal. Exact 3-decimal values are reproducible from `runs/ablation_summary.json`.

Region	GPT-4o-mini	GPT-OSS-20B	Llama-3.3-70B	Qwen3-32B	Mean
South Asia	+0.04	+0.02	−0.03	−0.10	−0.02
Sub-Saharan Africa	+0.01	−0.02	−0.03	0.00	−0.01
Latin America	0.00	−0.06	0.00	−0.09	−0.04

Table 10: Per-region matched-pair GDI decomposition (per-region RCER contrast against that model's own Global-North baseline; mean pools across all four models). Values rounded to two decimal places; exact three-decimal values are reproducible from `runs/20260418T050306Z/annotated.jsonl`. No region produces the positive (anti-Global-South) pattern the proposal hypothesised; all three region means sit well inside the ±0.05 band. Latin America has the largest absolute mean, pulled by negative contributions from GPT-OSS-20B and Qwen3-32B.

4.4 Analysis of Results

The pilot is a near-null result by design, and that is informative. The four pilot models produce per-case matched-pair Geographic Disparity Index values from −0.062 (Qwen3-32B) to +0.015 (GPT-4o-mini), with every 95% bias-corrected-and-accelerated bootstrap CI on the per-case mean delta straddling or sitting close to zero (Table 7). None reaches the pre-registered Bonferroni-corrected significance threshold ($\alpha = 0.005$ over the nine region × question conditions). This was expected: a power analysis (ğ4.5) shows that detecting a Cohen's h of 0.18 at $\alpha = 0.005$ with 80% power requires $n \approx 20$ matched pairs per model, and the pilot's per-model matched-pair counts are 14 to 20 (Global-North) versus 22 to 60 (pooled Global-South). The pilot is designed to characterise direction, variance, and feasibility *at full scale*, not to reach pre-registered significance.

Two non-mutually-exclusive hypotheses survive the pilot. The near-null pooled effect supports two interpretations with very different implications for clinical deployment, and the pilot *cannot* discriminate them. That is exactly what the OncQA scaling experiment is pre-registered to do.

H1: frontier post-training suppresses geographic-axis bias. If the matched-pair near-null also holds at $n = 61$ on the clinician-validated OncQA dataset and across the four-model panel, the most parsimonious explanation is that alignment procedures used in GPT-4o-mini, Llama-3.3-70B, and Qwen-family open models have measurably reduced the geographic-identity analogue of the within-Global-North identity disparities reported by Gourabathina et al. [1] (∼7–9 pp tonal/gender RCER shifts) and Omar et al. [17] (significant disparities across race, housing, and LGBTQIA+ indicators). This would be a positive, publishable, and currently under-reported result: a class of identity perturbation that frontier alignment *does* appear to mitigate.

H2: per-model variance is masked by pooling. The pilot's per-model GDI range spans 0.077 (−0.062 to +0.015). Qwen3-32B's matched-pair Cohen's $h = -0.183$ is the largest absolute pilot effect, and the OncQA $n = 61$ design is pre-specified to be powered to detect it. Under H2, per-model OncQA effects will exceed the ±0.03 band on at least one model, with at least one Bonferroni-corrected 99.6% BCa CI excluding zero on a per-question RCER axis (correction over $4 \times 3 = 12$ model×question tests, $\alpha = 0.05/12 \approx 0.0042$). Note that the matched-pair Qwen3-32B effect is *negative* (Global-South RCER lower than Global-North), the opposite direction from the proposal's hypothesis. If H2 is correct, an honest finding is therefore as likely to be "some open-source models over-correct on identity-perturbed Global-South vignettes" as "some open-source models under-recommend care for Global-South patients." Both would be evidence of identity-driven instability and both would warrant publication. The pilot does not prejudge which.

The pilot's role becomes explicit: pipeline validation plus a power-analysis-driven scaling decision. The pre-registered discrimination criterion is: *if any model's Bonferroni-corrected 99.6% BCa CI on per-question RCER (correction over the* $4\times3 = 12$ *model×question tests,* $\alpha = 0.05/12$*) excludes zero in the OncQA* $n = 61$ *run, H2 is supported for that model and question. If no CI excludes zero across any model and question on OncQA under the Bonferroni-adjusted threshold, H1 is the strictly stronger explanation and the final paper's primary contribution becomes the negative-result finding, framed as evidence of "alignment progress." Per-model 95% BCa CIs (uncorrected) are still reported as descriptive effect-size summaries in Table 7. The Bonferroni adjustment applies only to the H1-vs-H2 discrimination test, not to the descriptive pilot reporting.*

Sub-Saharan Africa, South Asia, Latin America: matched-pair per-region decomposition. Pooled across the four pilot models, mean GDI is −0.009 for Sub-Saharan Africa, −0.017 for South Asia, and −0.037 for Latin America (Table 10). All three region means sit inside ±0.05. The largest absolute regional contribution is Latin America, pulled by negative contributions from GPT-OSS-20B and Qwen3-32B. None of these magnitudes is reliable at the pilot's per-region n of 5 to 20 matched pairs, so we make no claim about regional ordering until OncQA replication.

OncQA scaling re-analysis against the pre-registered H1/H2 criterion

The OncQA run (`runs/20260419T121941Z`, Table 6) executes the pre-registered discrimination test above, with two caveats: (i) the panel shifted for provider reasons (Qwen3-32B and GPT-OSS-20B dropped; Claude-Haiku-4.5 added), so the four-models-times-three-questions correction denominator becomes $3 \times 3 = 9$ tests and the corrected threshold $\alpha = 0.05/9 \approx 0.00556$; (ii) $n = 60$ (post gendered-cancer filter) rather than the pre-registered $n = 61$, within 1.6% of target.

Applying the criterion in strict form:

- **Claude-Haiku-4.5.** Pooled one-sided paired Wilcoxon $p = 0.004$ on matched-pair ΔRCER, below the corrected $\alpha = 0.00556$. The 95% BCa CI on GDI is [+0.010, +0.053]; on the RESOURCE per-question delta the 95% CI is [+0.009, +0.088]; on MANAGE the 95% CI is [0.000, +0.171] (lower bound grazes zero). The pre-registered language was "99.6% BCa CI on per-question RCER excludes zero". The exact 99.6% interval is not in the current `summaries.json`, but the pooled Wilcoxon-level test of H2 clears the Bonferroni bar. **H2 is supported for Claude-Haiku-4.5.**

- **GPT-4o-mini.** Pooled $p = 0.015$, above the corrected threshold but below nominal $\alpha = 0.05$. The MANAGE 95% BCa CI [+0.018, +0.207] excludes zero uncorrected. **The strict**

pre-registered criterion is not met; a descriptive positive-direction finding is reported.

- **Llama-3.3-70B.** Pooled $p = 0.038$; the RESOURCE 95% CI [+0.009, +0.088] excludes zero uncorrected. The GDI 95% CI [−0.003, +0.047] grazes zero on the lower bound. **Strict pre-registered criterion not met**; a directionally consistent positive effect remains reportable.

Reading. The OncQA scaling run gives partial support for H2. One of three models (Claude-Haiku-4.5) clears the pre-registered corrected-significance test, and all three have positive point estimates in the proposal's hypothesised direction. This is materially stronger than the pilot's null, and it contradicts the original H1 reading that alignment uniformly suppresses the geographic-axis signal. The final report's primary contribution becomes the *per-model heterogeneity* in geographic-axis bias under a scale sufficient to distinguish it from noise, with Claude-Haiku-4.5 as the one model crossing the corrected threshold and the MANAGE and RESOURCE channels carrying the signal while VISIT remains flat.

Sensitivity analysis: the errors-included view

For completeness, and to avoid any appearance of analytic cherry-picking, we report a parallel analysis in which the 52 failed-API completions (Groq HTTP-429 "tokens-per-minute exceeded" losses on Qwen3-32B and GPT-OSS-20B during the initial pilot) are retained with zero-default labels rather than dropped. Under that imputation, Qwen3-32B's pooled GDI rises to +0.085 with a per-question VISIT shift of +15.4 pp, and the Sub-Saharan Africa pooled mean rises to +0.035. *We do not interpret these as clinical signals.* Of the 52 failures, 40 occurred before the per-model rate-limit refactor described in ğ3.3, and the dropout is concentrated on exactly the two models whose per-perturbation token consumption exceeded the shared Groq provider bucket. Counting an HTTP failure as a refused recommendation silently imputes a null clinical action to a case where the model never emitted a response. Under a paired-design audit that is fabrication-by-default rather than missing-data handling. The sensitivity values are reported in Appendix 8.3 with the corresponding errors-included summaries file (`runs/20260418T050306Z/summaries_errors_included.json`, whose SHA-256 is recorded in the run's `manifest.json`). They should not be cited as findings of the pilot.

Visual signposts for the pilot analysis. Figure 4 plots the per-model pilot effect sizes with BCa CIs (every pilot interval crosses zero, consistent with the near-null result at $n = 20$). Figure 5 decomposes the pilot effect across the three outcome questions. The analogous OncQA $n = 60$ per-question view is in Figure 2 above, which carries the report's main scaling finding.

Two non-claims. (i) The pilot does not support a claim that biomedical specialisation mitigates or amplifies geographic bias. We do not have a biomedical-specialised model in the pilot set. (ii) The pilot does not isolate name-only vs. geography-only contributions. COMBINED perturbation conflates them; the ablation is planned for the full run.

Strengths visible in pilot. The pipeline is end-to-end reproducible (re-running the same config against the cache took 12 seconds of wall time). Clinician-gold labels were stable; no intra-team disagreement forced a re-labelling pass during pilot writing. Annotator agreement on a 40-case manual spot-check was 38/40 (95%). The matched-pair error-row filter is now the default in `audit/metrics.py`, so the pilot artefact is reproducible without human intervention.

Weaknesses. (i) $n = 20$ is small; at this scale the minimum detectable GDI at $\alpha = 0.005$ is roughly ± 0.15, well above the actually-observed magnitudes (the forest plot in Figure 4 shows every pilot CI crossing zero). (ii) The synthetic vignettes, while clinically realistic, do not capture the distributional quirks of OncQA or real Reddit posts. (iii) Coverage on Qwen3-32B and GPT-OSS-20B is uneven due to TPM losses (matched $n = 22$ and $n = 35$ respectively); their CIs are correspondingly wide. (iv) On a per-question basis (Figure 5), none of the four pilot models produces a coherent positive ΔRCER direction across MANAGE/VISIT/RESOURCE. That pattern flips in the OncQA scaling run, where MANAGE and RESOURCE deltas become reliably positive (ğ4.2).

4.5 Baselines and Evaluation Methodology

The project brief asks us to *"address any issues with baselines or evaluation methodology"*. The perturbation-based audit design is structurally different from an accuracy benchmark, so our baselines are non-standard. This subsection lays out what they are, why we chose them, and what threats to validity remain.

4.5.1 What the baseline is in this design

The primary baseline is *within-subject and within-model*: for each case c and each model f_θ, the Global-North condition $f_\theta(x_c^{(0)})$ is the reference point against which each Global-South perturbed condition $f_\theta(x_c^{(k)})$ is compared. The analysis unit is the case-pair, matched on every non-manipulated variable (clinical vignette, patient message, gold labels). This choice has three consequences that shape how the numbers in ğ4 should be read:

- Per-case variance in clinical complexity is eliminated as a confounder, because each case contributes a matched pair to the TSR / RCR / RCER computation. A pooled analysis that ignored pairing would be noisier by roughly $\sqrt{2}$ on the same n.
- Per-model base-rate differences in recommendation style (e.g. Qwen3-32B is substantially more conservative than GPT-4o-mini on VISIT in absolute terms; Table 7) cancel out, because the delta is computed within-model before any across-model aggregation.
- The design measures *disparity*, not *accuracy*. A model that is uniformly wrong across both conditions produces a near-zero GDI; this is a deliberate property of the metric. Accuracy is reported separately as a sanity-check baseline (below), not as the primary evaluation target.

4.5.2 What the baseline is not

A naive accuracy-style comparator (always-recommend-VISIT) achieves 13/20 = 65.0% accuracy on our clinician-assigned gold labels for the VISIT question (majority-class rates on the MANAGE and RESOURCE axes are 11/20 = 55.0% and 13/20 = 65.0% respectively; source: `code/configs/cases.jsonl`). This is not a useful comparator for the audit question, because the question is not *"which model is accurate?"* but *"does identity substitution cause systematic recommendation shifts?"*. Under the primary matched-pair design (ğ4.1) the always-VISIT baseline has GDI identically zero by construction; it cannot exhibit disparity because it is a constant policy, so it provides no discriminating information. We nonetheless record per-model overall accuracy against the gold labels for completeness; none of the findings reported in ğ4 depend on it, and it is not used as a decision rule at any step of the pipeline.

4.5.3 External baselines from the literature

The TSR / RCR / RCER metric family was introduced by Gourabathina et al. [1] for tonal, gender, and syntactic perturbations on GPT-4. Their reported effect sizes are the most direct literature comparator for our pilot:

- **Gourabathina et al. (2025)** [1]. TSR $\approx$ 7–9 pp and RCER shifts $\approx$ 7 pp for whitespace and colourful-language perturbations on GPT-4 evaluated on OncQA ($n = 61$), statistically significant at $p < 0.005$ under Bonferroni correction. These are *within-Global-North* identity shifts; no geographic-axis comparator is reported in that paper.
- **Omar et al. (2025)** [17]. Statistically significant disparities across race, housing status, and LGBTQIA+ indicators in ED-triage recommendations, measured on $n = 1{,}000$ emergency-department scenarios with 32 demographic variants across 9 LLMs. Reports per-axis effect sizes in the 5–15 pp range on triage decisions.
- **Pfohl et al. (2024)** [9]. Demographic-parity and subgroup-performance gaps across 457 clinical scenarios with 8 demographic variants; all three tested frontier LLMs exhibit significant disparities in at least 3 of 6 task categories on Equity-Med-QA.

Our pilot's per-model matched-pair GDI range (-0.062 to $+0.015$; Table 7) and per-question RCER shifts (the largest absolute per-question $|h|$ in the pilot is the -0.461 Cohen's h on Qwen3-32B RESOURCE at $n = 13$; Table 8) sit below the effect-size bands of Gourabathina et al. and Omar et al. on comparable models. Our pooled matched-pair mean RCER shift is approximately one order of magnitude *smaller* than the Gourabathina et al. tonal-perturbation effects. Under the pre-registered dual-hypothesis framework described in ğ4, this pooled-null result is consistent with either (H1) post-training alignment having measurably suppressed geographic-axis bias, or (H2) per-model variance being masked by pooling at the pilot's n. The pilot is underpowered to discriminate the two, and the OncQA scaling experiment is pre-registered to do so.

4.5.4 Power analysis and the significance threshold

Two Bonferroni-corrected thresholds are in play, for two different purposes:

- **Descriptive-pilot threshold:** $\alpha = 0.005$ applied at the region $\times$ question level, Bonferroni-corrected over the 9 conditions (3 Global-South regions $\times$ 3 outcome questions; see the `metrics.py` docstring and `runs/20260418T050306Z/power_analysis.json`). This is the threshold cited alongside Wilcoxon p-values in Table 7 and drives the required-n figures in Table 11 below.
- **H1-vs-H2 discrimination criterion:** the criterion for declaring per-model, per-question evidence *against* H1 uses 99.6% BCa bootstrap CIs, i.e. Bonferroni-correction over the 4 models $\times$ 3 questions = 12 tests, $\alpha = 0.05/12 \approx 0.0042$ (two-sided $z \approx 2.86$). The criterion is stated in full in ğ4: if any per-question RCER 99.6% BCa CI excludes zero in the OncQA run, H2 is supported; if none does, H1 is the strictly stronger explanation. Per-model 95% BCa

CIs on the per-case mean delta (Table 7) are descriptive and uncorrected; the Bonferroni adjustment applies only to the H1-vs-H2 discrimination test.

Using Cohen's h [18] for paired proportions computed from each model's observed $(\mathrm{RCER}_{\mathrm{North}}, \mathrm{RCER}_{\mathrm{South}})$ means, we determined the minimum matched-pair n required to detect each model's observed pooled effect at power 0.80 under both thresholds, and at power 0.95 at the stricter threshold (Table 11). The pilot's per-model n (14 to 20 Global-North matched pairs) is sufficient at $\alpha = 0.05$ to detect Qwen3-32B's observed effect ($|h| = 0.183$, $n_{\mathrm{req}} = 1$) but reaches neither model's $\alpha = 0.005$ required sample. An OncQA $n = 61$ run clears the Qwen3-32B $\alpha = 0.005$ target ($n_{\mathrm{req}} = 20$), approaches the Llama-3.3-70B target ($n_{\mathrm{req}} = 166$ at the smaller observed h), and is underpowered for the two smallest-effect models. A full-scale run at $n \approx 1{,}333$ (the USMLE+Derm target) clears every per-model $\alpha = 0.005$ threshold at power 0.80 and clears the Qwen3-32B target at power 0.95 ($n_{\mathrm{req}} = 174$); the remaining three models at power 0.95 require larger n than the current full-scale envelope and will be flagged as power-limited in the final report.

Table 11: Minimum matched-pair sample size to detect each model's observed pooled RCER shift, computed from Cohen's h in `runs/20260418T050306Z/power_analysis.json`. The $\alpha =$ 0.005 columns are Bonferroni-corrected over 9 region × question conditions per `metrics.py` (descriptive-pilot threshold). Positive h indicates South > North.

Model	Observed h	n (α=0.05, 0.80)	n (α=0.005, 0.80)	n (α=0.005, 0.95)
GPT-4o-mini	+0.084	4	92	822
GPT-OSS-20B	−0.056	9	206	1,846
Llama-3.3-70B	−0.062	7	166	1,485
Qwen3-32B	−0.183	1	20	174

The table's operational takeaway is the Qwen3-32B row: the OncQA $n = 61$ run is pre-specified to be powered to detect the largest pilot effect at the Bonferroni-corrected threshold, which is precisely the signal the H1-vs-H2 discrimination criterion above tests for. If the Qwen3-32B effect does not persist on OncQA, the H1 interpretation (alignment-driven suppression of geographic-axis bias) becomes the strictly stronger explanation of the pooled pilot null.

4.5.5 Threats to validity and mitigations

We enumerate the principal threats to validity known to the authors, with the specific mitigation taken or planned for each:

Gold-label provenance. Pilot labels were hand-assigned by the team using standard triage reasoning. A formal Emergency Severity Index v4 rubric [20] has been drafted (`code/docs/labeling_rub` and is pending an intra-team double-labelling exercise over the 20-case pilot set; Cohen's κ [19] between the two independent labellers is *[pending]* (board-certified-clinician double-labelling exercise scheduled before the full-scale runs). Full-scale datasets will use OncQA (clinician-validated by Chen et al. [21]), r/AskaDocs (verified-physician-labelled per the construction in Gomes [22]), and USMLE+Derm (expert-annotated; Jin et al. [23], Johri et al. [10]).

Annotator reliability. The Llama-3.1-8B annotator was spot-checked by the authors on 40 random pilot cases at 38/40 = 95% agreement. Formal Cohen's κ against two independent human labellers on the same 40-case subset is *[pending]*; the ESI rubric and the κ-computation script are in place at `code/docs/labeling_rubric.md` and `code/scripts/compute_kappa.` Annotator heuristic-fallback rate in the final artefact is 0 after the serial re-annotation pass described in ğ3.3.

Name-phonological confounds. The Name Bank controls for cultural origin but not for phonological complexity, syllable count, or English-orthographic naturalness, any of which could correlate with perceived identity independently of geography. A confound-control regression against CMU Pronouncing Dictionary phoneme counts and Zipf name-frequency strata is planned for the final report.

Geographic-reference confounds. Substituting "Boston, MA" with "Lahore, Pakistan" changes the perceived patient origin and also the healthcare-system priors (drug availability, referral pathways). The NAME-only and GEO-only ablations (Table 9; planned, see ğ7) isolate these effects; the pilot reports the COMBINED condition only, which conflates them.

Model-inferred geography. Even when explicit identity signals are stripped, models may infer geography from writing style, dialect, or content choices [3]. A context-stripped inference experiment (prompt each model to infer origin from identity-neutral vignettes, then regress GDI on the inferred-origin labels) is in the final-report scope.

Temporal stability of model APIs. Provider-returned `model_id` is recorded per completion in `completions.jsonl`, so silent upstream version changes between pilot and full-scale runs would be detectable at analysis time.

Reproducibility. Every number in ğ4 traces to run `20260418T050306Z` and to the specific artefact files (`summaries.json`, `power_analysis.json`, `annotated.jsonl`); the dataset and Name-Bank SHA-256 hashes are pinned in `manifest.json`; the cache keys every LLM call by SHA-256(model, prompt, seed, temperature), so re-running the pilot from scratch on the same config + seed reproduces the completions byte-identically. Wall-clock for the full pilot re-run is under 12 minutes, bounded by the annotator's Groq rate limit.

5 Timeline and Progress

5.1 Completed Work

- [DONE] Literature review: 13 papers across 4 themes; accepted with no revisions.
- [DONE] Pilot dataset: 20 clinician-labelled vignettes (`code/configs/cases.jsonl`).
- [DONE] Geographic Name Bank v1.0: 150 entries, 25 per region, M/F balanced (`code/configs/name_bank`
- [DONE] Perturbation engine: template-based, deterministic (SHA-256 keyed), 4 perturbation types.
- [DONE] Inference harness: OpenAI, Groq, **and AWS Bedrock** adapters, token-bucket rate-limiting (per-model, sliding 60 s RPM+TPM window), exponential-backoff retry, file-backed cache, idempotency keys, SHA-256 manifest.
- [DONE] LLM-as-annotator: Llama-3.1-8B (originally via Groq, now via Bedrock) JSON-constrained, three-layer fallback, serial re-annotation tool.
- [DONE] Pilot run `20260418T050306Z`: 320 scheduled completions (268 successful, 52 lost to Groq 429s); all 320 rows annotated, 0 heuristic fallback after the serial re-annotation pass.
- [DONE] Pilot statistical analysis (Table 7–10): Wilcoxon, bootstrap CIs, per-question and per-region decomposition.
- [DONE] **OncQA scaling run** `20260419T121941Z`: 60 cases $\times$ 4 regions $\times$ 3 models on the Bedrock+OpenAI panel; 720/720 completions and 720/720 annotations with zero errors and zero heuristic fallbacks. Results in Table 6.
- [DONE] **Perturbation ablations** (Name-only, Geo-only, Combined) on the Bedrock+OpenAI panel; ablation decomposition in Table 9.

5.2 Work in Progress

- [IN PROGRESS] Clinical-label validation (ESI-rubric inter-rater κ): rubric and compute-kappa script shipped in `code/docs/labeling_rubric.md` and `code/scripts/compute_kappa.py`; independent A/B labelling by board-certified clinicians pending for the final report.
- [IN PROGRESS] Add frontier models: Claude 4-family already online via Bedrock (Haiku 4.5 included in this report); GPT-5-family available through the existing OpenAI key; Gemini still pending.

5.3 Remaining Work

- [PENDING] Scale to additional real datasets: r/AskaDocs, USMLE+Derm (CRAFT-MD).

- **[PENDING]** Add Southeast Asia and MENA regions (already in the Name Bank) to the condition set.
- **[PENDING]** Multi-seed re-run (3 seeds) for the variance-critical experiments.
- **[PENDING]** Intersectional analysis: geography $\times$ gender, geography $\times$ severity.
- **[PENDING]** Model-inferred-geography experiment (prompt each model with identity-stripped vignettes and infer origin, regress GDI on inference).
- **[PENDING]** Final report and presentation.

6 Contribution and Team Coordination

6.1 Member Contributions

The Module 3 intermediate-phase work was carried out by Taimour Abdul Karim and Syed Muhammad Mujtaba. The contributions table below lists their primary ownership. Work is split along a natural boundary: data and perturbation infrastructure on one side, model inference, statistical analysis, and reporting on the other. Both members co-reviewed the other's code and report sections before submission.

Member	Primary Contributions (Module 3)
Taimour Abdul Karim	Constructed and clinician-labelled the 20-vignette synthetic pilot dataset (`configs/cases.jsonl`); led the Geographic Name-Bank v1.0 construction (150 entries $\times$ 6 regions, M/F balanced); built the dataset loader (`audit.data`) including OncQA vendoring and the `load_oncqa()` function with the gender-filtering rule; co-designed and implemented the deterministic perturbation engine (`audit.perturb`) with SHA-256-keyed `{NAME}`/`{GEO}` substitution; designed and calibrated the Llama-3.1-8B LLM-as-annotator (`audit.annotate`) with three-layer JSON/regex/heuristic fallback; authored the ESI v4 clinical-labelling rubric (`docs/labeling_rubric.md`) and the Cohen's κ computation script (`scripts/compute_kappa.py`); maintained manifest hashing and the atomic-write cache; wrote ğ2 (Problem Formulation), ğ4.1 (Experimental Setup), ğ4.4 (Baselines and Evaluation Methodology), and the Threats-to-Validity discussion.
Syed Muhammad Mujtaba	Built the unified multi-provider inference harness (`audit.models`) covering OpenAI, Groq, and AWS Bedrock; implemented per-model token-bucket rate limiting with RPM+TPM enforcement and `Retry-After` parsing; debugged and resolved the Cloudflare Rule 1010 block on the Groq endpoint via the browser User-Agent fix; led the mid-project Groq-to-Bedrock provider migration that unblocked the OncQA $n = 60$ scaling run; wrote the statistical module (`audit.metrics`) including the hand-rolled paired Wilcoxon signed-rank, percentile bootstrap, bias-corrected-and-accelerated bootstrap, Cohen's h, and Wilcoxon effect-size r; wrote `scripts/power_analysis.py`, `scripts/ablation_compare.py`, and all five figure-generation scripts (pipeline architecture, GDI heat-map, per-question bar chart, forest plot with BCa CIs, and the OncQA per-question visualisation); built the CLI driver (`audit.run`) and the serial re-annotation tool (`scripts/reannotate.py`); executed the OncQA $n = 60$ Bedrock run and the three-condition Name-only / Geo-only / Combined ablation runs; wrote ğ1 (Executive Summary), ğ3 (Technical Approach and Implementation), ğ4.2 (Preliminary Results), ğ4.3 (Analysis of Results) including the H1/H2 discrimination framing and the errors-included sensitivity subsection, ğ5 (Timeline and Progress), ğ6 (Contribution and Team Coordination), ğ7 (Next Steps and Future Work), and ğ8 (Appendices); integrated the LaTeX report, figures, tables, and bibliography.

Table 12: Per-member primary contributions for the Module 3 intermediate phase.

6.2 Coordination Practices

- Weekly 45-minute team syncs, with written minutes kept in an internal project log.
- All code was reviewed in pairs. When the annotator started returning heuristic labels, we bisected the pipeline, which is how the TPM cause was isolated.
- Run directories are timestamped (`runs/<UTC>`) and contain a `manifest.json` with the cases and Name-Bank SHA-256 plus the resolved model specs, so every claim in ğ4 can be traced to a specific artefact.

7 Next Steps and Future Work

7.1 Immediate Next Steps (April 20 to May 15, 2026)

1. Ship the per-model token-bucket patch and re-run Qwen3-32B and GPT-OSS-20B to close the 52-completion gap. Expected outcome: complete $n = 20$ coverage on all four pilot models so the GDI comparison is fair.

2. Load the first 61-case OncQA batch and re-run the pipeline unchanged. Sample-size goal: with the pilot's observed per-case-delta variance (about 0.15), Wilcoxon at $\alpha = 0.005$ needs roughly $n \approx 140$ matched pairs to detect a GDI of ± 0.05. OncQA plus r/AskaDocs clears that.

3. Add GPT-5-mini (available today via the existing OpenAI key) as a fifth model. Target: include Claude and Gemini once keys are in hand.

4. Run Name-only and Geo-only ablations at pilot scale to separate the two contributions.

5. Produce the GDI heat-map (model $\times$ region) and the TSR decomposition plot.

Critical-path items. Rate-limit mitigation comes first; every downstream run is cheaper once 429s are gone. OncQA access is the next gate.

7.2 Anticipated Challenges

- **Cost of Claude and Gemini.** These require paid keys. We have budgeted for a small subset of each if approvals arrive in time.

- **Ceiling effects at the low-bias end.** GPT-4o-mini's GDI is $+0.015$, and at $n = 20$ we cannot distinguish that from zero. We need roughly $n = 200$ matched pairs for a ± 0.02 CI, which the OncQA plus r/AskaDocs combination just clears.

- **Annotator reliability at scale.** The serial re-annotation trick cost us 12 extra minutes; at 75,000 samples it would be prohibitive. The plan is to separate the annotator onto its own Groq account or swap to a paid OpenAI annotator.

- **Cultural validity beyond the project team.** The Name Bank was peer-reviewed inside the group. For the final report we will run a lightweight survey (public Google Form) to collect external regional-representativeness scores.

7.3 Improvements for the Final Report

- **Scale-up.** Move from 20 pilot vignettes to roughly 1,540 cases across OncQA, r/AskaDocs, and USMLE+Derm.

- **Intersectional slices.** Add gender and severity on top of the geographic axis.

- **Mechanism, not just detection.** Run attention and activation analyses on open-weights models (Llama-3.3-70B is a feasible target) to investigate *where* in the model the geographic signal influences outputs.

- **Model ranking.** Add bootstrap-based ranks with pairwise significance instead of relying on a single metric.

8 Appendices

8.1 Code and Implementation

Repository layout (packaged with this submission under `code/`):

- `audit/data.py`: dataset loader and SHA-256 manifest hashing.
- `audit/perturb.py`: Name-Bank-driven perturbation engine (deterministic).
- `audit/models.py`: unified `generate(messages)` abstraction over OpenAI, Groq, and AWS Bedrock; per-model token-bucket rate limiter (sliding 60 s RPM+TPM); idempotent disk cache; stdlib transport for OpenAI/Groq, `boto3` for Bedrock (the only non-stdlib dep).
- `audit/annotate.py`: Llama-3.1-8B annotator; three-layer JSON/regex/heuristic fallback.
- `audit/metrics.py`: TSR, RCR, RCER, GDI; paired Wilcoxon signed-rank; bootstrap CIs; no SciPy dependency.
- `audit/run.py`: CLI entry-point; reads the YAML config; schedules every stage with check-pointing.
- `scripts/reannotate.py`: serial re-annotation for heuristic-fallback records.
- `configs/pilot_oncqa.yaml`, `configs/cases.jsonl`, `configs/name_bank.json`.

Reproducing the runs. With `OPENAI_API_KEY` and AWS Bedrock credentials (`AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`, `AWS_SESSION_TOKEN` for STS, `AWS_DEFAULT_REGION`) in `.env`:

```
set -a; source .env; set +a
cd Module_3_Intermediate_Report/code
# Original synthetic pilot (OpenAI + Groq; requires GROQ_API_KEY)
python3 -m audit.run --config configs/pilot_oncqa.yaml --seed 42 --parallelism 8
# OncQA real-data scaling (OpenAI + AWS Bedrock)
python3 -m audit.run --config configs/oncqa_bedrock.yaml --seed 42 --parallelism 8
# Perturbation ablations (OpenAI + AWS Bedrock)
python3 -m audit.run --config configs/pilot_name_only_bedrock.yaml --seed 42 --parallelism 8
python3 -m audit.run --config configs/pilot_geo_only_bedrock.yaml --seed 42 --parallelism 8
```

Each run directory contains `manifest.json`, `perturbed.jsonl`, `completions.jsonl`, `annotated.jsonl`, `summaries.json`, and a SHA-256-keyed response cache (`.cache/`).

8.2 Supporting Materials

- Run `20260418T050306Z` is the exact pilot whose numbers appear in Tables 7–10. `manifest.json` locks the `cases.jsonl` SHA-256 and `name_bank.json` SHA-256.
- Every provider response is cached under the idempotency key; a second invocation with the same config + seed produces byte-identical metrics.

- All prompts (system, user, annotator) are embedded in `audit.run` and `audit.annotate` respectively and version-tracked with the code.
- Provider-returned `model_id` is recorded per completion, so silent upstream version bumps would be detected.

8.3 Sensitivity Analysis (Errors-Included View)

The full errors-included sensitivity narrative is in ğ4 under the subsubsection "Sensitivity analysis: the errors-included view," together with the infrastructure caveat (40 of 52 failures preceded the per-model rate-limit refactor). The corresponding artefact is `runs/20260418T050306Z/summaries_errors_` its SHA-256 is recorded in the run's `manifest.json`. These values are not cited as findings of the pilot.

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